

**Generative AI Applications: Conditional Generation of HKDSE ICT Exam Questions
Using a Fine-Tuned Transformer**

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MSc in Artificial Intelligence — Capstone Project 5: Generative AI Applications

[8 October 2026]

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Overview

This report describes the design, implementation, and evaluation of a conditional text generation system that produces novel multiple-choice questions (MCQs) in the style of the Hong Kong Diploma of Secondary Education (HKDSE) Information and Communication Technology (ICT) Paper 1A examination. The system addresses a practical need in Hong Kong secondary education: producing candidate questions for review and practice without replicating existing examination material verbatim.

The chosen generative approach is Transformer-based causal language modelling (Vaswani et al., 2017). An autoregressive decoder model generates text token by token, conditioned on a header that specifies the curriculum module (A–E) and question-type identifier. This approach is appropriate for the task because generating a question stem with four answer options is fundamentally a text-sequence completion problem, which is the setting that decoder-only Transformers are designed for. Generative Adversarial Networks and Variational Autoencoders are primarily used for continuous data such as images and are not standard tools for variable-length structured text generation (Goodfellow et al., 2014; Kingma & Welling, 2014).

The base model is TinyLlama-1.1B-Chat-v1.0 (Zhang et al., 2024), a 1.1-billion-parameter LLaMA-architecture decoder pre-trained on general English text. It was fine-tuned using Low-Rank Adaptation (LoRA; Hu et al., 2022), a parameter-efficient method that attaches small trainable matrices to the attention layers while keeping the base weights frozen. Training was performed on a Kaggle notebook with two NVIDIA Tesla T4 GPUs.

Dataset Description

The dataset consists of 560 HKDSE ICT Paper 1A multiple-choice questions from the 2012–2025 examination series (14 papers $\times$ 40 questions), published by the Hong Kong Examinations and Assessment Authority (HKEAA). It is available at <https://www.kaggle.com/datasets/stansir/dataset/data>. The dataset was constructed by converting the official paper PDFs to Markdown and structuring one CSV row per question, with the question stem, four answer options (A–D), the official answer key, image references where a figure accompanies the question, and the cohort percentage from each year's Candidate Performance report.

Each question was classified against the topic taxonomy of the HKEAA ICT Curriculum and Assessment Guide (Secondary 4–6, effective 2022/23), assigning a question-type identifier (qn_type_id) that encodes the curriculum module (A–E) and subtopic. Five compulsory modules are represented: Module A — Information Processing (154 questions), Module B — Computer System Fundamentals (84), Module C — Internet and Its Applications (119), Module D — Computational Thinking and Programming (74), and Module E — Social Implications (50). A further 79 questions carry the label “unclassified” because the classification work for those questions has not yet been completed; they are included in the training data to maximise the available examples. Full field definitions are in the companion file `fields_explanation.md`.

Three questions whose answer options consist entirely of images (the [IMAGE] sentinel) were excluded because their options cannot be represented as text, leaving 557 usable questions. After preprocessing, a stratified 80/20 split by module produced 445 training examples and 112 validation examples. The questions were written by HKEAA and are used here for non-commercial academic study only. The dataset was not reused from any previous capstone project and was not synthetically generated.

The dataset is appropriate for the generative task because it is the complete population of examination questions from the target domain, it is publicly available, and it was created by human subject-matter experts, which is the kind of output the model is trained to produce. Its main limitation as a training corpus is size: 445 examples is small for fine-tuning a language model, and every question was produced under the same house style, which limits the variety the model can learn from.

Model Design and Training Approach

TinyLlama-1.1B-Chat-v1.0 was chosen as the base model for three reasons. First, it is a decoder-only Transformer that performs autoregressive text generation natively, which matches the task of writing a question and four options as a continuation of a conditioning header. Second, at 1.1 billion parameters it is substantially more capable than smaller alternatives such as GPT-2 (Radford et al., 2019) while still fitting on a single free GPU without quantisation. Third, it has been pre-trained on a large English corpus and instruction-tuned, so it already possesses general vocabulary, grammar, and some knowledge of ICT topics; fine-tuning only needs to teach the examination format and DSE-specific content style.

LoRA (Hu et al., 2022) was selected as the fine-tuning method rather than full parameter updates because the training set is small (445 examples). Full fine-tuning of 1.1 billion parameters on such a dataset would almost certainly overfit and erase the pre-trained knowledge the model needs for vocabulary and coherence. LoRA inserts trainable low-rank matrices (rank $r = 16$, scaling factor $\alpha = 32$, dropout 0.05) into the four attention projections of every decoder layer (q_proj, k_proj, v_proj, o_proj), adding 4,505,600 trainable parameters, which is 0.41% of the total. The base weights remain frozen throughout training.

Every training example is formatted as a conditioning header followed by the question and options, ending with the end-of-sequence (EOS) token: "Module: [name] / Type:

[qn_type_id] // Question: [stem] // Options: / [A–D options][EOS]". The model is trained with causal language-modelling cross-entropy loss computed over the complete example, including the header, so it learns both the content and the layout. A key implementation detail is that the padding token was set to the tokeniser's unknown token (<unk>) rather than the EOS token; using EOS as padding would mask the real EOS from the loss and prevent the model from learning to stop.

Before formatting, the question stems were cleaned to remove artefacts from PDF extraction: question numbers (e.g. **12.**), Markdown image links replaced by plain-text placeholders ([Figure: caption]), shared-passage text and paper footers ("END OF PAPER") that appeared after option D in a subset of questions. A maximum sequence length of 384 tokens was chosen after inspecting the token-length distribution; no example exceeded this threshold.

Training used the Hugging Face Trainer with AdamW (Loshchilov & Hutter, 2019), learning rate 2×10^{-4} , cosine schedule with 20 warm-up steps, weight decay 0.01, and an effective batch size of 16 (4 per device $\times$ 2 gradient-accumulation steps $\times$ 2 GPUs) for 5 epochs. FP16 mixed precision was enabled. Training loss fell from 2.09 to 0.88 and validation loss fell from 1.11 at epoch 1 to 0.957 at epoch 4, then stayed flat at epoch 5 (perplexity 2.6), indicating stable convergence with no over-fitting.

Output Evaluation and Interpretation

To evaluate generation quality, the fine-tuned model and the untuned base model (LoRA adapter switched off) were each prompted with 10 question-type headers, drawn as the two most frequent classified types per module from the training set, and asked to generate 2 samples each, producing 20 samples per model. Three supporting metrics were computed: format validity (stem and exactly four options A–D present), module consistency (whether a TF-IDF nearest-centroid classifier, 82% accurate on real validation questions, placed the

sample in the prompted module), and copy ratio (share of 6-word sequences that also appear in a real question). Additionally, the first sample per prompt was read qualitatively.

The fine-tuned model produced a valid format (stem + four options) in 18 of 20 samples (90%), against 0 of 20 for the base model. The base model failed completely: for the a_ip_03 prompt it wrote a physics question-and-answer exchange about classical and quantum fields; for a_ip_04 it wrote a course-module description of a computer vision pipeline; for b_cs_02 it wrote an essay on memory buses. None of its 20 samples constitutes a multiple-choice question. This contrast demonstrates that fine-tuning was necessary for the task, not merely beneficial.

No sample from either model was a near-copy of a real question (0 of 20 with a copy ratio ≥ 0.5 ; mean copy ratio for the fine-tuned model was 0.055). The fine-tuned model reproduced DSE examination conventions including the multi-statement format — listing statements (1)–(3) followed by combination options such as "A. (1) and (2) only ... D. (1), (2) and (3)" — and scenario stems such as "Mary decides to use social media to send out invitations to a dinner party..." (e_si_04). Of the 10 printed samples, only e_si_04 was judged usable without editing, because it has one clearly defensible answer (option B).

Three categories of failure were observed. First, generation control: two samples hit the 260-token budget without reaching four options, one repeating " $n \leftarrow n + 1 / n \leftarrow n - 1$ " in an infinite loop and one cutting off an SQL question mid-option. Second, incoherence: the a_ip_03 sample asked about "the functions of the following symbols in data transfer" and listed "(1) S (2) 8 (3) 10 (4) 12" with options "File format / File size / File type / Storage"; the symbols bear no relationship to the options. Third, assessment design: several samples had overlapping distractors (c_ia_02: "secure password", "strong password", "keep the password safe"), no fully correct option (c_ia_01: "Encapsulation" and "Encoding" are not browser functions), or an answer given away (b_cs_04 asks for a queue size of 100 and offers "100"

as option B). Module consistency was 75% for the fine-tuned model against 60% for the base model, a small gap for a proxy that is itself 82% accurate, indicating only weak module conditioning.

The distinct-2 score (unique word-pair share across all samples) was 0.62 for the fine-tuned model, compared to 0.45 for the 557 real questions; however, this comparison is not directly valid because it was computed on 20 versus 557 texts, and shorter lists produce higher distinct-n. The 10 printed stems nonetheless cover varied topics — browser functions, cloud storage, switch statements, social media, SQL queries — so the model does not collapse onto a single template.

Ethical Considerations and Responsible Use

The most immediate ethical concern is misuse of generated content as authoritative examination material. HKDSE questions are set by subject-matter experts who calibrate difficulty, verify factual accuracy, pilot-test distractors, and ensure one unambiguous correct answer. The model was not trained on any of that quality-assurance information: no answer rationale, no difficulty calibration, and no distractor-design principles appear in the training text. As shown in the evaluation, only 1 of 10 printed samples was usable without editing, and four samples had no single defensible answer. Deploying the system without expert review could expose students to incorrect or misleading content, which is a direct educational harm (Bender et al., 2021).

A second concern is curriculum bias and topical imbalance. Module A (Information Processing) contributes 154 training examples while Module E (Social Implications) contributes only 50. The model has seen approximately three times as many Module A questions as Module E questions. This imbalance is an inherent property of the HKEAA examination design — Module E carries fewer hours and fewer examination questions than Module A — but it means the model is less reliably conditioned for Module E prompts, and

any generated Module E questions may reflect the majority-module style rather than Module E conventions. More broadly, the model was trained exclusively on English-language questions from one subject in one examination system; it has no knowledge of other curricula, other languages, or other subjects, and generalising its outputs beyond HKDSE ICT would be unjustified.

A third concern is copyright and creative ownership. The questions were written by HKEAA and remain HKEAA copyright material. The model was fine-tuned on this material for non-commercial academic study, which is a standard fair-use context in Hong Kong and most jurisdictions; however, generating and distributing large numbers of questions that closely imitate HKEAA style could be considered an infringement of the intellectual property that underpins the examination system. The near-copy check (0 samples above the 0.5 threshold) suggests the model is not reproducing verbatim passages, but style imitation is not explicitly excluded by standard copyright doctrine.

To use the system responsibly, all generated questions should be reviewed by a qualified ICT teacher before use, the answer key should be independently verified, and generated content should be clearly labelled as AI-drafted rather than HKEAA-produced. The system is appropriate as a drafting aid to reduce the time a human expert spends producing candidate questions, not as a replacement for human examination-setting.

Limitations and Future Improvements

The most significant limitation is dataset size. With 445 training examples the model is learning from a small sample of the possible question space for five curriculum modules. This restricts the variety of stems and distractors it can produce and makes it susceptible to memorising the surface patterns of the training questions rather than learning deeper principles of question construction. A larger corpus — additional subjects, additional years,

or questions from comparable curricula in other jurisdictions — would substantially reduce this problem.

The second limitation is the absence of figure-aware generation. Of the 557 usable questions, 145 (26%) reference a figure that the text model cannot see. The model learned that many DSE questions reference figures and reproduces [Figure: ...] placeholders, but it cannot generate or describe the figure contents. This means a human expert must supply every figure, which limits the system's usefulness for questions in modules where figures are common, particularly Modules A and D. Integrating a vision-language model — for example, extending the approach of Liu et al. (2024) with LLaVA — to jointly generate question text and a corresponding diagram would address this limitation.

The third limitation is the absence of answer-key generation and distractor quality control. The model produces four options but does not indicate which is correct and does not enforce that exactly one option is correct and the others are wrong for identifiable reasons. A natural extension would be to include the answer key in training — appending "Answer: [A/B/C/D]" after the options — and to add a post-generation verification step that checks each distractor against a knowledge base or a second model prompted to identify correct answers.

The fourth limitation is weak module conditioning. The 75% module-consistency score for the fine-tuned model against 60% for the base model shows that the module name in the header provides some conditioning signal but not a strong one. Stronger conditioning techniques such as prefix tuning (Li & Liang, 2021) or classifier-free guidance (Ho & Salimans, 2022) applied to a text model may improve topic adherence without requiring more training data.

Future work should also address evaluation methodology. The metrics used here — format validity, TF-IDF module consistency, copy ratio, and distinct-2 — are heuristics that

measure form rather than educational quality. A more rigorous evaluation would involve ICT teachers rating generated questions on difficulty appropriateness, factual accuracy, and distractor quality, following established item-analysis procedures (Haladyna & Rodriguez, 2013). A student-facing pilot study would reveal whether questions produced by the system have the intended diagnostic value.

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